Type I and Type II Errors, Power, and One- vs Two-tailed Tests

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Hypothesis testing: null and alternative.
- **Compare with:** Confusion matrix.

1. Overview

Key point: Every test decision can be wrong in two ways: rejecting a true H_0 (Type I) or failing to reject a false H_0 (Type II); the direction of H_1 decides whether the rejection region sits in one tail or two.

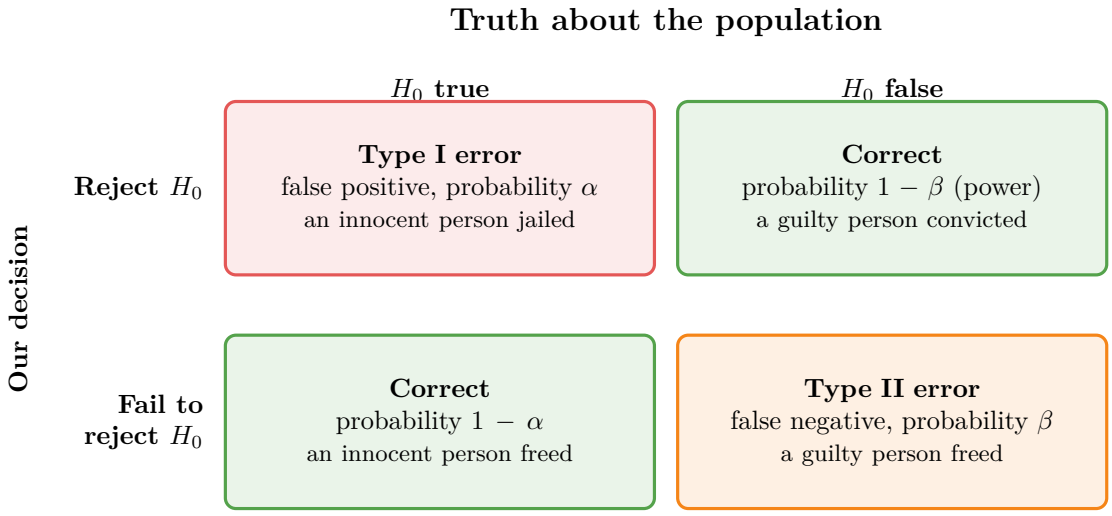


Figure 1: The four possible outcomes of a hypothesis test

A hypothesis test ends with one of two decisions, reject H_0 or fail to reject it (see the eight steps of a hypothesis test). The truth is also one of two: H_0 true or H_0 false. Figure 1 crosses them into four outcomes, two of which are errors.

This Note covers:

- Type I and **Type II errors** (G-2033), with their probabilities α (alpha, the significance level we choose in advance) and β (beta, the chance of missing a real effect);
- the **power of a test** (G-1539), $1 - \beta$;
- the trade-off between the two errors;
- one-tailed and **two-tailed tests** (G-2028), and the pros and cons of each;
- where hypothesis testing is used, in general and in machine learning.

2. Type I and Type II errors

> **Key point:** A **Type I error** (G-2032) rejects a true H_0 (a **false positive** (G-748), probability α); a **Type II error** (G-2033) fails to reject a false H_0 (a **false negative** (G-747), probability β).

Whatever we decide, the decision may be wrong:

- we say "reject H_0", but H_0 was actually true;
- we say "fail to reject H_0" for lack of evidence, but H_0 was actually false.

The other two outcomes are correct: rejecting a false H_0, and failing to reject a true one.

Figure 2 runs the training-program test of the z-test example ($H_0 : \mu = 50$, $H_1 : \mu > 50$, $\sigma = 5$, $n = 30$) 200 times in each of two worlds. Each dot is the **z statistic** of one test (how many standard errors the sample mean lies above 50); the **critical value** 1.645 is the boundary beyond which we reject H_0 (see the rejection region and the critical value):

- H_0 **true (left):** 13 of the 200 tests land beyond 1.645 and reject a true H_0 (red), close to the 5 percent that α promises;
- H_0 **false, true mean 52 (right):** 61 of the 200 tests stay below 1.645 and miss the real effect (orange), close to the 29 percent computed in section 3.

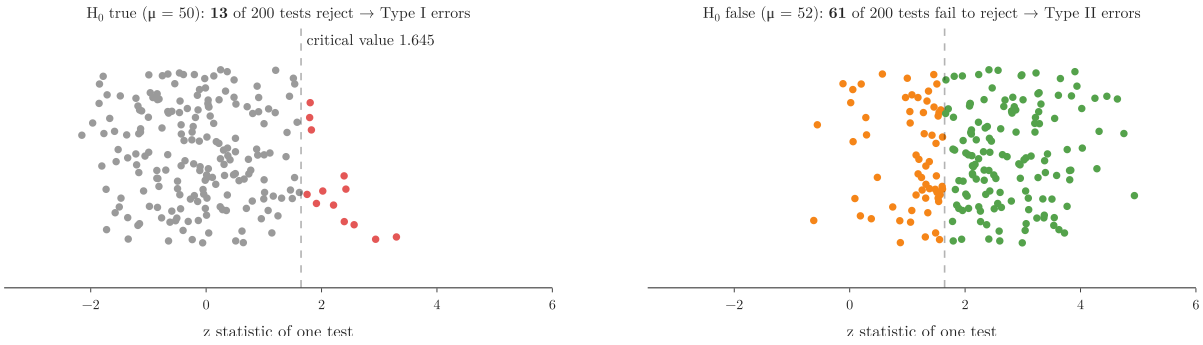


Figure 2: 200 simulated tests when H0 is true (left) and 200 when the true mean is 52 (right). Red: Type I errors, beyond the critical value 1.645. Orange: Type II errors, below it. Green: correct rejections.

2.1 Type I error

A **Type I error** (G-2032) occurs when the sample leads us to reject H_0 although it is in fact true. A Type I error is the mistake of finding a significant effect or relationship when there is none, also called a **false positive** (G-748).

In the courtroom of the three rules for H_0, H_0 says "no crime". A Type I error convicts an innocent person.

The probability of a Type I error is the significance level α (see how rare is too rare). The

researcher chooses α, so this risk is **under our control**. Lowering α from 5% to 1% pushes the critical values out and leaves more room for the innocent to go free.

2.2 Type II error

A **Type II error** (G-2033) occurs when the sample leads us to fail to reject H_0 although it is in fact false. The researcher fails to detect an effect or relationship that actually exists. A Type II error is also called a **false negative** (G-747).

In the courtroom: the accused did commit the crime, but the evidence was too weak and they walk free.

The probability of a Type II error is written β.

2.3 The error table

	H_0 true	H_0 false
Reject H_0	Type I error (probability α)	correct (probability $1-\beta$)
Fail to reject H_0	correct (probability $1-\alpha$)	Type II error (probability β)

The second row is sometimes labelled "accept H_0". As failing to reject H_0 showed, the correct wording is "fail to reject H_0": the test never proves H_0.

The same two errors appear in classification. In the **confusion matrix** (G-449) of the confusion matrix, a false positive is a **Type I error** (G-2032) and a false negative a Type II error. The link is direct: "positive" means "we flag an effect", which in a test is "reject H_0".

3. Power of a test

> **Key point:** The power, $1-\beta$, is the probability that the test rejects H_0 when H_0 is false: its ability to detect a real effect.

The **power of a test** (G-1539) is $1-\beta$: the probability of correctly rejecting a false H_0. A powerful test seldom misses a real effect. In practice, power matters more than β itself.

β and the power depend on how false H_0 is. Figure 3 works this out for the training program of the z-test example ($H_0 : \mu = 50$, $H_1 : \mu > 50$, $\sigma = 5$, $n = 30$). In each panel the horizontal axis is the z statistic. The blue bell is where it falls if H_0 is true (mean 50); the dashed bell is where it falls if the true mean is 52. The black line is the critical value. The red area (blue bell beyond the line) is α, the orange area (dashed bell before the line) is β, and the green area (dashed bell beyond the line) is the power.

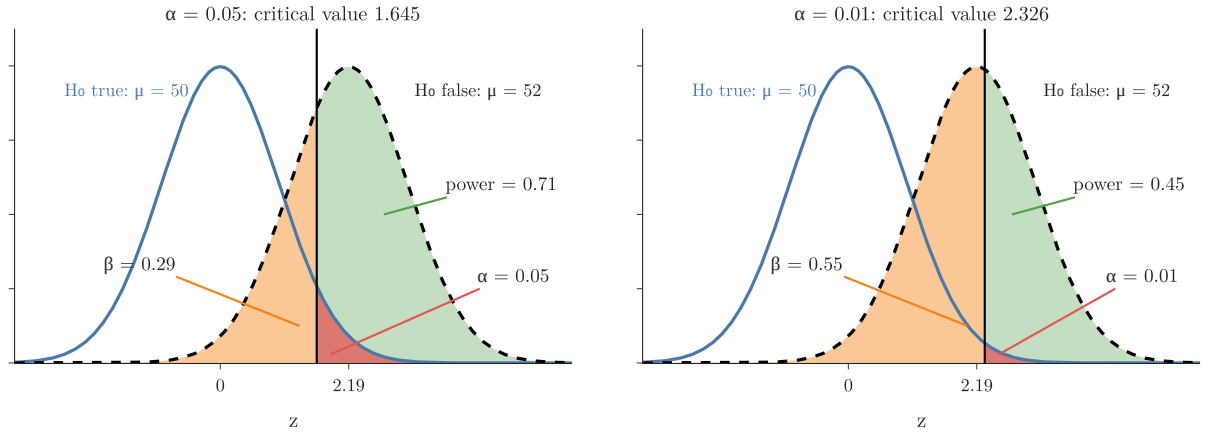


Figure 3: Type I error (red), Type II error (orange) and power (green) for the training-program test if the true mean is 52

Extra: Computing the power, assuming the true mean is 52.

1. **In words:** if $\mu = 52$, the z statistic is no longer centred at 0 but at the true difference in standard errors. Power is the chance that it still lands beyond the critical value.
2. **Formula:** with shift d and right-tailed critical value z_α,

$$d = \frac{\mu_{\text{true}} - \mu_0}{\sigma/\sqrt{n}}$$

$$\beta = \Phi(z_\alpha - d)$$

$$\text{power} = 1 - \beta$$

   where Φ is the standard normal CDF.
3. **Example:** the shift is

$$d = \frac{52 - 50}{5/\sqrt{30}} = 2.19$$

   At $\alpha = 0.05$:

$$\beta = \Phi(1.645 - 2.19)$$

$$\beta = \Phi(-0.55) = 0.29$$

$$\text{power} = 1 - 0.29 = 0.71$$

   So even if the training really added 2 cars a day, this test would miss it 29% of the time.

4. The trade-off between Type I and Type II errors

> **Key point:** For a fixed sample, lowering α shrinks the rejection region and so raises β: we cannot push both errors down at once.

Lowering α protects against **Type I errors** (G-2032), but it has a price. A smaller α widens the region where we fail to reject H_0. A future test statistic is then more likely to land there, including when H_0 is false: more guilty people walk free.

Figure 3 shows it with numbers. For the same true mean of 52:

α	Critical value	β	Power
0.05	1.645	0.29	0.71
0.01	2.326	0.55	0.45

Cutting α from 5% to 1% almost doubles the chance of missing the effect. The two errors move in opposite directions, so we strike a balance, much as a smoke alarm is set sensitive enough to catch fires but not so sensitive that it sounds for every slice of toast. The usual setting, 0.05, is such a compromise, unless a domain expert has a specific reason to change it.

> **Extra:** The way to lower both errors is more data. Keeping $\alpha = 0.05$ and the true mean at 52, the power of the training test rises from 0.71 with 30 employees to 0.93 with 60 and 0.99 with 100. A larger sample shrinks the standard error, so the two curves of Figure 3 move apart. Less variable data does the same thing: with $\sigma = 4$ instead of 5 (and $n = 30$), the standard error is smaller and the power rises from 0.71 to 0.86.

Figure 4 turns the three knobs one at a time, using the power formula of section 3. Watch the bars on the right: lowering α moves the critical line right and swaps power for β; a bigger sample or a bigger true effect pushes the dashed curve away from H_0, and power climbs while α stays at 0.05.

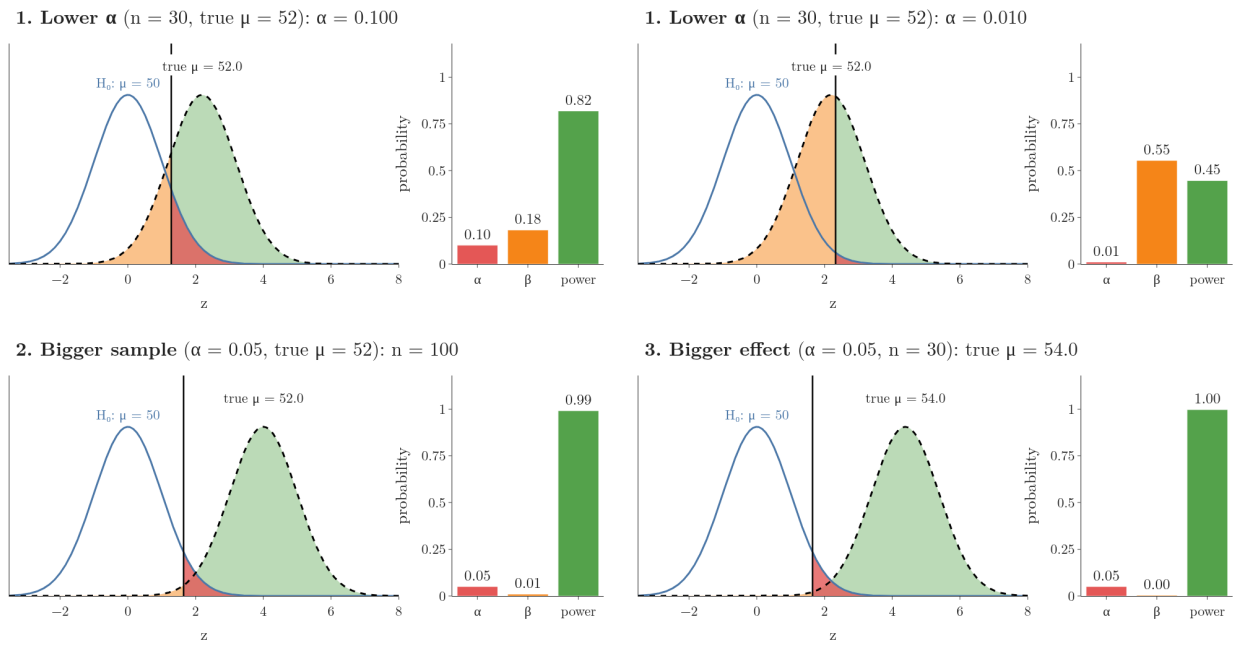


Figure 4: The three knobs of a test's power, for the training-program test. 1: α from 0.10 to 0.01 (β grows from 0.18 to 0.55). 2: n from 10 to 100 (power reaches 0.99). 3: the true mean from 50 to 54 (at 50 the power equals α). Red = α, orange = β, green = power

5. One-tailed and two-tailed tests

> **Key point:** H_1 with $>$ or $<$ gives a one-tailed test with all of α in one tail; H_1 with $\neq$ gives a two-tailed test with $\alpha/2$ in each tail.

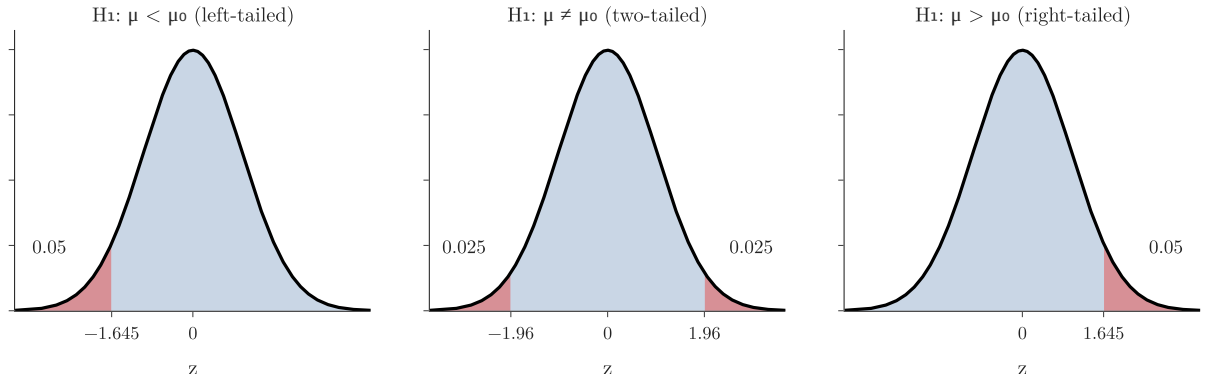


Figure 5: Rejection regions (red) at $\alpha = 0.05$ for the three kinds of alternative hypothesis

The form of H_1 fixes where the rejection region goes (Figure 5).

- A **one-tailed test** (G-1385), or one-sided test, is used when we test for an effect in one specific direction. H_1 contains an inequality sign:

  - **right-tailed**, $H_1 : \mu > \mu_0$: "the new filming style increases the mean view duration"; "a new medication increases the average recovery rate compared with the existing one";
  - **left-tailed**, $H_1 : \mu < \mu_0$: "the new style decreases the mean view duration".
- A **two-tailed test** (G-2028), or two-sided test, is used when we test for an effect in either direction. H_1 contains $\neq$: "the chips packets do not weigh 100 g", more or less.

The rule of thumb: an H_1 that says "increases" or "decreases" is one-tailed; an H_1 that says "changes" or "differs" is two-tailed. At $\alpha = 0.05$ the critical values are 1.645 (or -1.645) for one tail and ± 1.96 for two (see one tail or two).

5.1 Two-tailed tests: pros and cons

Advantages:

- **They detect effects in both directions.** Wherever z falls, far left or far right, the test can see it. They suit situations where the direction is uncertain, or where we want to know about any difference between groups.
- **They are more conservative in each direction.** Each tail gets only $\alpha/2$, so a result in a given direction must be more extreme (beyond 1.96 instead of 1.645) to count.

Disadvantages:

- **They are less powerful** for an effect in a known direction. Because α is split, a larger effect is needed to reject H_0. For the training program with a true mean of 52, a **two-tailed test** (G-2028) at 5% has power 0.59 against 0.71 for the right-tailed test.
- **They are not designed for a directional hypothesis.** The question "is the weight different from 100 g?" is not "is it lower?".

> **Extra:** The overall **Type I error** (G-2032) rate of a two-tailed test is still α, only split across two tails. And when a two-tailed test rejects H_0, the sign of z tells the direction: $z = -2.5$ means the mean is below 100 g.

5.2 One-tailed tests: pros and cons

Advantages:

- **They are more powerful in the chosen direction.** All of α sits in one tail, so the rejection region on that side is larger and a real effect there is easier to detect.
- **They fit a directional hypothesis.** When there is a strong theoretical or practical reason to expect one direction, such as a training program that can only help, the **one-tailed test** (G-1385) asks exactly that question.

Disadvantages:

- **They miss effects in the other direction.** If the new filming style actually lowers the view duration, a right-tailed test cannot see it: z lands far in the left tail, which is not a rejection region, and we "fail to reject" a null that is wrong in the opposite way.
- **They are easy to misuse.** The direction must be fixed before seeing the data.

Extra: The real Type I risk of one-tailed tests comes from choosing the tail after looking. With a correct one-tailed test the **Type I error** (G-2032) rate is exactly α. But if we look at z first and then test in the direction it points to, we reject whenever $|z| > 1.645$: in a simulation of 100,000 tests with a true H_0, that happens 10.0% of the time, double the promised 5%.

6. Where hypothesis testing is used

Key point: Testing interventions, comparing groups, finding relationships, checking distributions, testing independence and A/B testing; in ML, comparing models, selecting features, tuning and checking assumptions.

6.1 In general

1. **Testing the effect of an intervention or treatment.** Does a new filming style raise view time? Does a new drug work? With σ known we use a z-test, otherwise a t-test (see the t statistic step by step).
2. **Comparing means and proportions** between two or more groups: customer satisfaction scores, conversion rates (proportions, i.e. percentages), employee performance, the average marks of section A and section B. The tools are t-tests (see the independent two-sample t-test) and ANOVA (a test that compares the means of several groups at once).
3. **Relationships between variables.** Is the correlation between two numerical **features** (G-772; input variables, columns of the data table) real or chance? Pearson's correlation coefficient (see reading a correlation) and Spearman's rank correlation each come with a test.
4. **Goodness of fit.** Does a dataset follow a normal, binomial or Poisson distribution? The chi-square test (which compares counts we observed with the counts a distribution predicts) is one tool.
5. **Independence of categorical variables.** On the Titanic, are sex (male, female) and survival (1, 0) related, or independent? Is the type of product related to whether customers return it? The chi-square test answers this.
6. **A/B testing.** Two versions of a web page, thumbnail or product (A and B) are shown to random groups of users, and a test decides which converts or engages better (see the machine learning development life cycle). Tech companies run such tests constantly; A/B testing is hypothesis testing standardized for marketing, product development and website design.

6.2 In machine learning

1. **Model comparison.** We train XGBoost, a random forest and linear regression on the same data. Is one really better, or did it win by chance on this split? A **paired t-test** (G-1439) on the scores of the same **cross-validation** (G-510) folds (see cross-validation with a pipeline) answers it; comparing two machine learning models runs one.

2. **Feature selection.** Which features are related to the **target** (G-1949; the output we predict), and which can we drop (see **feature selection** (G-768))? A t-test, chi-square test or ANOVA between each feature and the target gives a p-value per feature. scikit-learn's `f_classif` computes the ANOVA F statistic (a score that is large when a feature's values differ between the classes) and its p-value (the chance of a result this extreme if H_0 were true) for each feature, and `chi2` the chi-square statistic; `SelectKBest` (G-1762) keeps the k features with the highest scores (scikit-learn §1.13). Figure 6 does this on scikit-learn's wine dataset (178 wines, 13 features, 3 classes): every feature's p-value is below 0.00001, so all are related to the class, and the F statistic ranks them; `SelectKBest(k=5)` keeps the five highest.

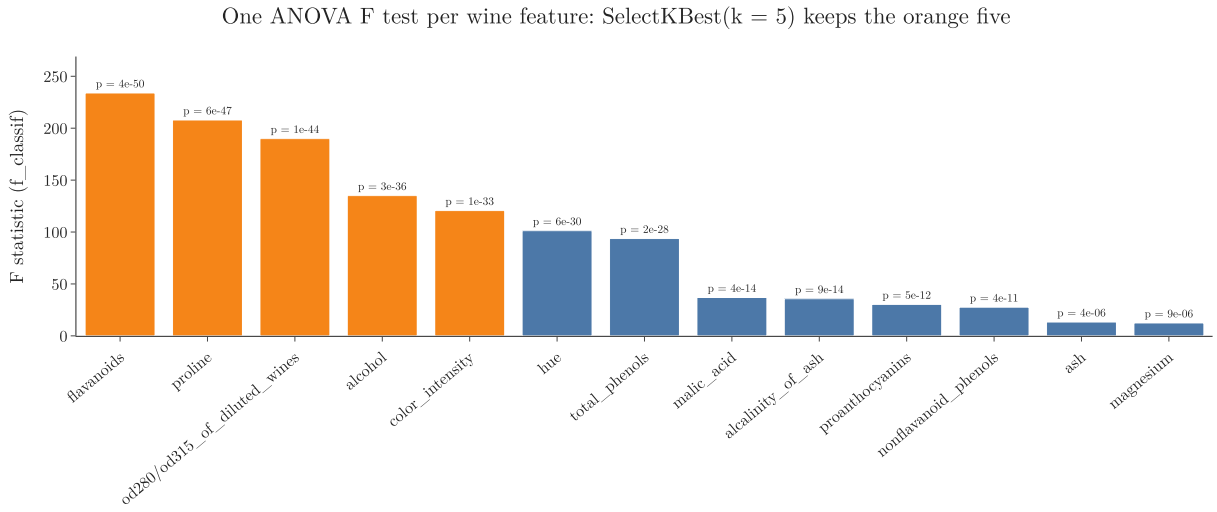


Figure 6: One ANOVA F test per feature of the wine dataset (f_classif), with each p-value above its bar. SelectKBest(k = 5) keeps the five features with the largest F (orange).

3. **Hyperparameter tuning.** Two settings of a model give different cross-validation scores (see one split is not enough). A test says whether one setting is significantly better.

4. **Checking model assumptions.** Linear regression assumes, among other things, normally distributed residuals (see normal residuals). The Shapiro-Wilk test checks normality (Shapiro and Wilk 1965); if the assumptions fail, the model may not suit the data.

Libraries such as scikit-learn usually run these tests for us. Knowing what happens behind the scenes lets us use those tools better and make better-educated decisions.

7. Summary

	One-tailed	Two-tailed
H_1	$\mu > \mu_0$ or $\mu < \mu_0$	$\mu \neq \mu_0$
Rejection region	all of α in one tail	$\alpha/2$ in each tail
Critical value, $\alpha = 0.05$	1.645 or -1.645	± 1.96
Strength	more power in the chosen direction	detects effects in both directions
Weakness	blind to the other direction	less power in each direction

- Use the table to choose: one-tailed only when there is a strong reason to expect one direction (a training program that can only help), two-tailed when the direction is uncertain.
- **Type I error** (G-2032): reject a true H_0 (**false positive** (G-748)), probability α, chosen by us, so this risk is under our control.
- **Type II error** (G-2033): fail to reject a false H_0 (**false negative** (G-747)), probability β, which is why "fail to reject" never proves H_0.
- Power is $1 - \beta$: the chance of detecting a real effect; 0.71 for the training test if $\mu = 52$, so even a real gain of 2 cars a day would be missed 29% of the time.
- For a fixed sample, lowering α raises β, because a smaller rejection region also catches fewer real effects; more data (or less variable data) lowers both, because it shrinks the standard error.
- Fix the direction of H_1 before looking at the data, because choosing the tail after looking doubles the real Type I error rate (10.0% instead of 5%).
- This answers the opening question: a decision can be wrong in two ways, Type I with probability α and Type II with probability β, and the direction of H_1 puts the rejection region in one tail or two.

8. Sources

Built from

- CampusX, "Session 45 - Hypothesis Testing Part 1 | DSMP 2023", YouTube, https://www.youtube.com/watch?v=S94mx6OL7kM
- Khan Academy, "Introduction to power in significance tests", YouTube, https://www.youtube.com/wat

Other references

- scikit-learn developers. *User Guide*, §1.13 "Feature selection" (univariate selection with `SelectKBest`, `f_classif`, `chi2`). scikit-learn.org, feature_selection.html.
- Shapiro, S. S. and Wilk, M. B. (1965). "An analysis of variance test for normality (complete samples)." *Biometrika* 52(3–4), 591–611.

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Type II error (false negative) (G-2033)	A miss: the test finds no effect when there really is one (failing to reject H_0 when it is actually false); its probability is β.
Power of a test (G-1539)	The probability that a test detects a real effect, that is, rejects H_0 when H_0 is false; it equals $1 - \beta$, where β is the probability of a Type II error.
Two-tailed test (two-sided test) (G-2028)	A test that looks for an effect in either direction: H_1 uses $\neq$, and the significance level is split into $\alpha/2$ in each tail.
Type I error (false positive) (G-2032)	A false alarm: the test declares an effect when there is none (rejecting H_0 when it is actually true); its probability is α.
One-tailed test (one-sided test) (G-1385)	A test whose H_1 has a direction ($>$ or $<$), with the whole rejection region in one tail.
β (G-5)	The chance that a test misses a real effect, failing to reject a false H_0 (the probability of a Type II error). $1 - \beta$ is the test's power.
False positive (FP) (G-748)	Predicted positive, but actually negative; a Type I error.
False negative (FN) (G-747)	Predicted negative, but actually positive; a Type II error.
Confusion matrix (G-449)	A table counting a classifier's predictions for every pair of actual and predicted class, so we can see which kinds of mistake it makes, which accuracy alone hides.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Paired t-test (dependent t-test) (G-1439)	A test of whether two linked measurements of the same subjects differ on average; it works on the difference within each pair (a t-test on the differences).
Cross-validation (G-510)	Testing a model by training and testing it several times on different parts of the training data.
Target (G-1949)	The output column we predict, such as the class; also called the label.
Feature selection (G-768)	Keeping only the useful input columns and dropping the rest.

Term	Meaning
SelectKBest (G-1762)	scikit-learn class that gives every column a score and keeps the `k` best, so only the most useful features go into the model.
Right-tailed and left-tailed test (G-1693)	One-tailed tests that count only one tail as evidence against H_0: a right-tailed test rejects for large values of the test statistic (as for $H_1 : \mu > \mu_0$), a left-tailed test for small ones (as for $H_1 : \mu < \mu_0$).